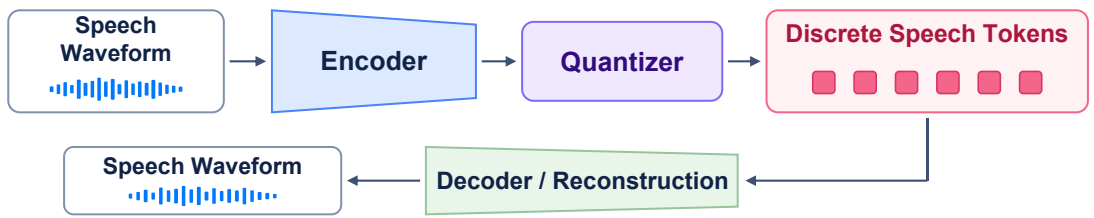
Speech Components: Discrete and Continuous Speech Representations

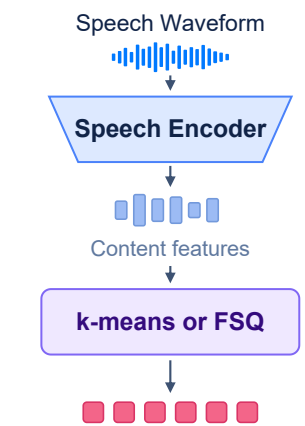
Overview of tokenization, continuous speech encoding, and speech decoding pathways

(a) Discrete Speech Representation

Speech Tokenizer

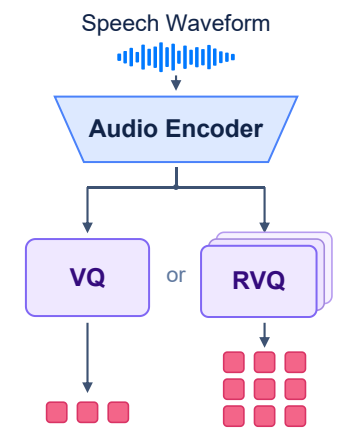


Semantic-oriented



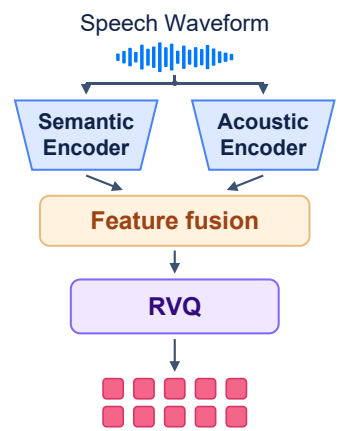
e.g., HuBERT, CosyVoice 2

Acoustic-oriented



e.g., SoundStream, EnCodec, WavTokenizer

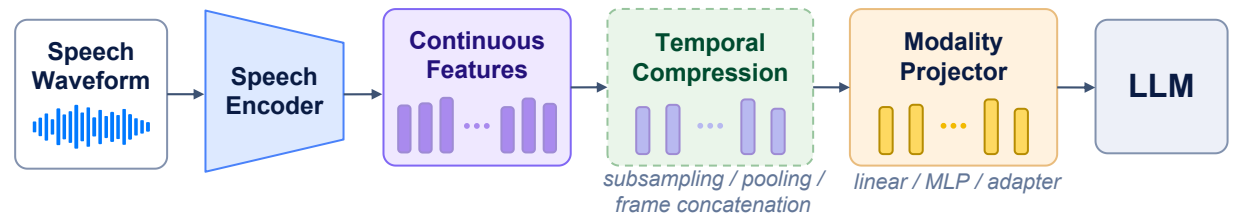
Semantic–acoustic hybrid



e.g., SpeechTokenizer, X-Codec, Mimi, DualCodec

(b) Continuous Speech Representation

Speech Encoder



Speech Decoder

